



Mutual Fund Analytics Dashboard

Capstone Project I – Mutual Fund Analytics

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Data Analyst Internship Program

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The **Mutual Fund Analytics Dashboard** is an end-to-end financial analytics platform developed using **Python, Pandas, SQLite, Scikit-Learn, Plotly**, and **Streamlit**.

The platform enables investors to:

- Explore mutual fund schemes
- Analyze performance metrics
- Assess investment risk
- Generate personalized recommendations



Data Analytics

Processing and analyzing large-scale mutual fund datasets



Financial Analytics

Calculating CAGR, Sharpe Ratio, and risk-adjusted returns



Dashboard Development

Interactive Streamlit dashboard with real-time filters



Fintech Applications

Practical fintech solution for investment decision support

Challenges Investors Face



Large Number of Available Schemes

Thousands of mutual fund schemes make selection overwhelming for investors



Complex Financial Metrics

NAV, expense ratio, Sharpe ratio, and other metrics are difficult to interpret



Manual Comparison is Time Consuming

Comparing funds across multiple parameters manually takes hours of research



Risk Assessment is Difficult

Evaluating volatility, drawdown, and VaR requires specialized knowledge

The Need

An **automated analytics system** capable of generating meaningful insights and personalized recommendations.

Such a system would enable investors to:

- Make data-driven investment decisions
- Compare funds efficiently
- Understand risk profiles
- Receive tailored recommendations



Collect Datasets
Gather mutual fund data from multiple sources



Clean & Validate
Process and validate financial data quality



Store in SQLite
Structured storage for efficient querying



Calculate Metrics
Compute CAGR, Sharpe, Sortino ratios



Analyze Risk
Volatility, drawdown, and VaR analysis



Generate Recommendations
Profile-based fund suggestions



Build Dashboard
Interactive Streamlit interface



Deploy on Cloud
Streamlit Cloud deployment

End-to-End Pipeline
From raw data collection to cloud deployment — a complete analytics workflow

8 Steps

#	Dataset Name	Description
1	Fund Master Dataset	Core fund information and scheme details
2	NAV History Dataset	Historical Net Asset Value records over time
3	AUM Dataset	Assets Under Management tracking data
4	Monthly SIP Inflows	Systematic Investment Plan contribution data
5	Category Inflows	Fund category-wise investment flow analysis
6	Investor Transactions	Individual investor transaction records
7	Portfolio Holdings	Fund portfolio composition and holdings
8	Benchmark Indices	Market benchmark data for comparison

All datasets were cleaned, validated, transformed and integrated before analytics processing.

Data Processing Pipeline

01 Data Collection from Sources



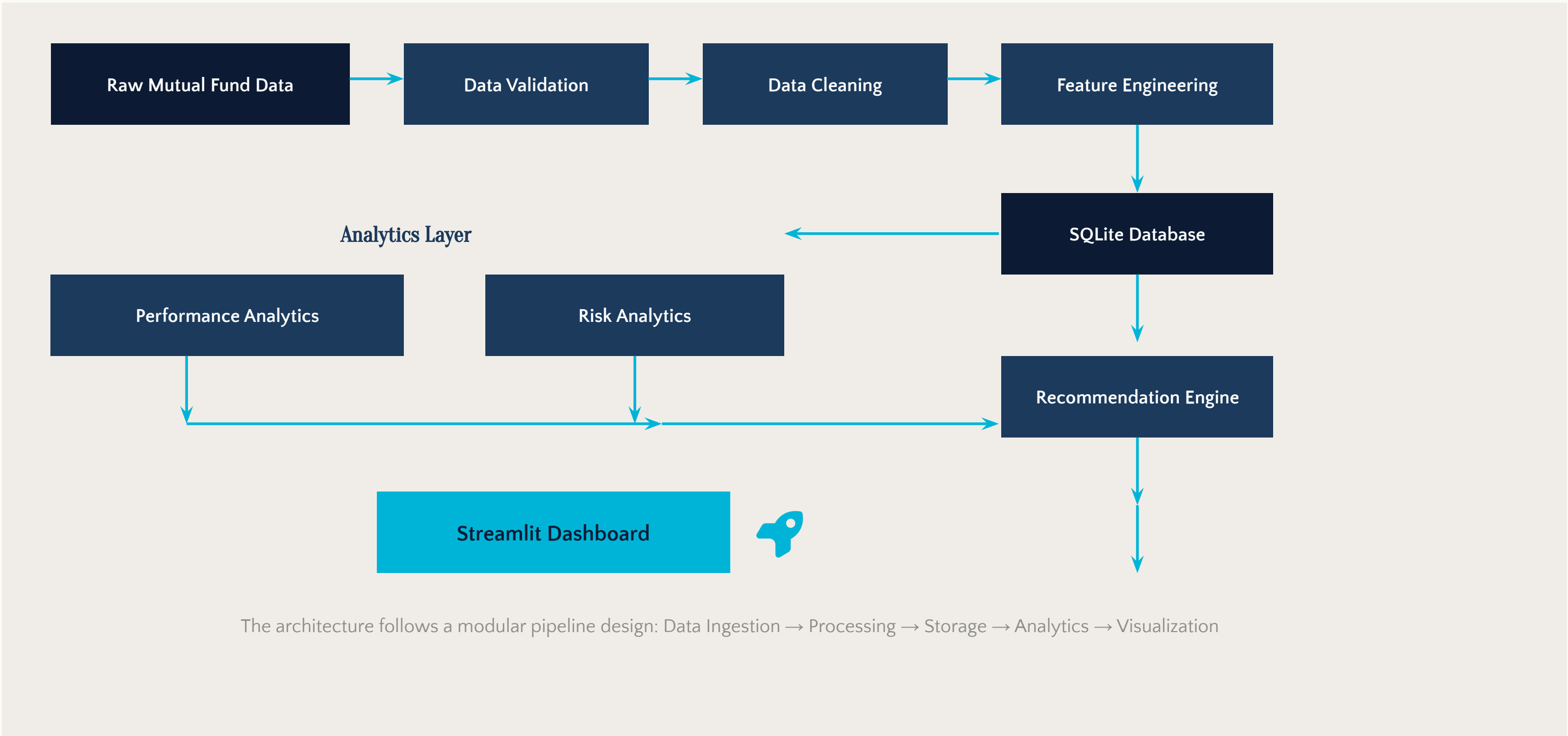
02 Data Cleaning & Validation



03 Data Transformation



04 Integration into SQLite



Programming

- Python

Data Processing

- Pandas
- NumPy

Database

- SQLite

Machine Learning

- Scikit-Learn

Visualization

- Plotly
- Matplotlib

Dashboard & Deployment

- Streamlit
- Git & GitHub
- Streamlit Cloud

Key Performance Metrics



CAGR

Compound Annual Growth Rate — measures annualized return over a multi-year period



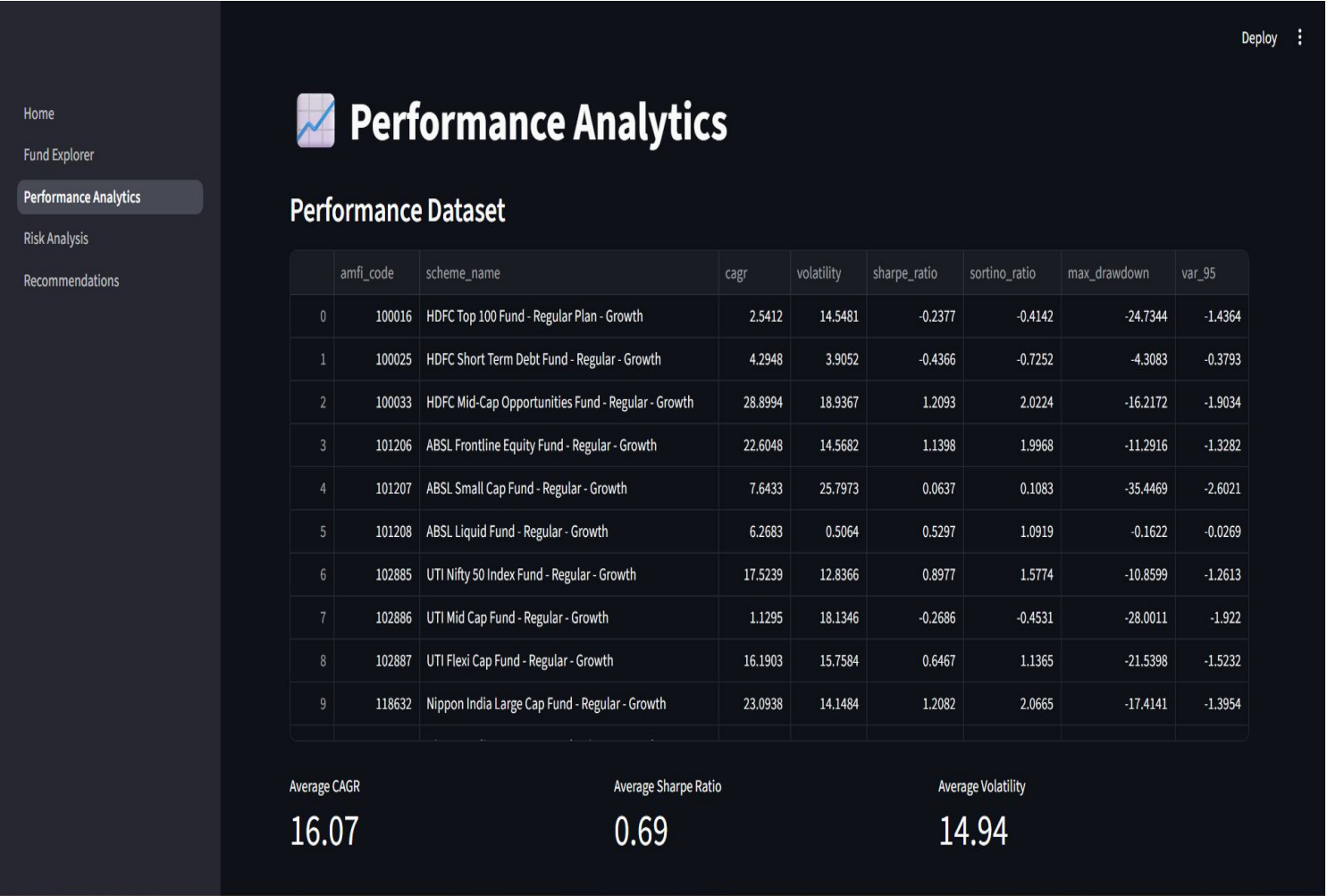
Sharpe Ratio

Risk-adjusted return metric comparing return to volatility — higher is better



Sortino Ratio

Downside risk-adjusted return — focuses only on harmful volatility



Insight: Performance ranking of mutual fund schemes based on historical returns and risk-adjusted performance metrics enables informed investment decisions.

Key Risk Metrics



Volatility

Standard deviation of returns — measures the degree of variation in fund performance over time



Maximum Drawdown

Largest peak-to-trough decline — indicates the worst-case scenario for investment loss



Value at Risk (VaR)

Estimates potential loss at a given confidence level — quantifies downside risk exposure



The recommendation engine uses performance and risk metrics to generate suitable fund recommendations based on investor profiles.

 Conservative

Prioritizes capital preservation with low-risk funds, debt-oriented schemes, and stable returns.

- Low Volatility
- Debt Funds
- Stable Returns

 Moderate

Balanced approach with hybrid funds, moderate risk tolerance, and diversified portfolio.

- Medium Volatility
- Hybrid Funds
- Balanced Risk-Return

 Aggressive

High growth focus with equity funds, higher risk tolerance, and long-term wealth creation.

- High Volatility
- Equity Funds
- High Growth Potential



How it works: The engine analyzes fund performance metrics (CAGR, Sharpe, Sortino) and risk indicators (volatility, drawdown, VaR) to match funds with investor risk profiles, generating personalized top fund recommendations.

>20%

Several funds achieved CAGR greater than 20%

Strong

Relationship observed between return and volatility

Ranked

Risk-adjusted rankings identified high quality funds

Profiled

Recommendation engine generated profile-based suggestions

Detailed Findings

- CAGR Analysis:** Multiple equity funds demonstrated CAGR exceeding 20% over a 5-year horizon, significantly outperforming benchmark indices
- Risk-Return Correlation:** Strong positive correlation observed between historical returns and volatility, validating modern portfolio theory assumptions
- Dashboard Impact:** The unified analytics dashboard simplified complex fund analysis from hours of manual research to minutes of interactive exploration





<https://mutualfundanalytics-a5cjfcqbqf5bb9bjydutnb.streamlit.app/>

Deployment Features



Cloud Hosted

Deployed on Streamlit Cloud with 24/7 availability



Interactive Filters

Dynamic filtering by category, risk, and returns



Real-Time Experience

Live dashboard updates with instant visual feedback



Download Functionality

Export analysis results and fund reports as CSV



Public Accessibility

Anyone can access the dashboard via web browser

- Dashboard is Live and Accessible
- | • Built with Streamlit + Python
- | • Deployed June 2026
- | • [Github Repository](#)

 Live Mutual Fund APIs

Real-time data feeds from AMFI and other financial data providers

 PostgreSQL Migration

Scalable database solution for handling larger datasets

 Portfolio Optimization

Modern Portfolio Theory based optimal allocation suggestions

 Monte Carlo Simulations

Probabilistic return forecasting using simulation models

 Advanced ML Recommendations

Collaborative filtering and content-based recommendation models

 Automated Reports

Scheduled PDF and email report generation for portfolios



Mobile Responsive Design

Fully responsive dashboard design optimized for mobile and tablet devices, enabling on-the-go fund analysis and portfolio monitoring.

The project successfully combines **data analytics**, **financial analytics**, **database management**, **machine learning**, and **dashboard development** into a unified mutual fund analytics solution.



Practical Fintech Application

Real-world financial analytics solution for investment decision-making



Data-Driven Decisions

Empowers investors with quantitative insights and recommendations



Successful Deployment

Live cloud deployment with public accessibility and interactive features

This capstone project demonstrates proficiency in the complete data analytics pipeline — from data collection and processing to machine learning and cloud deployment — delivering a production-ready fintech analytics solution.



Thank You

Questions & Discussion

Arun Vk

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